ICDFA | WADF - 2026-M01 | Linux Essentials | Week 02

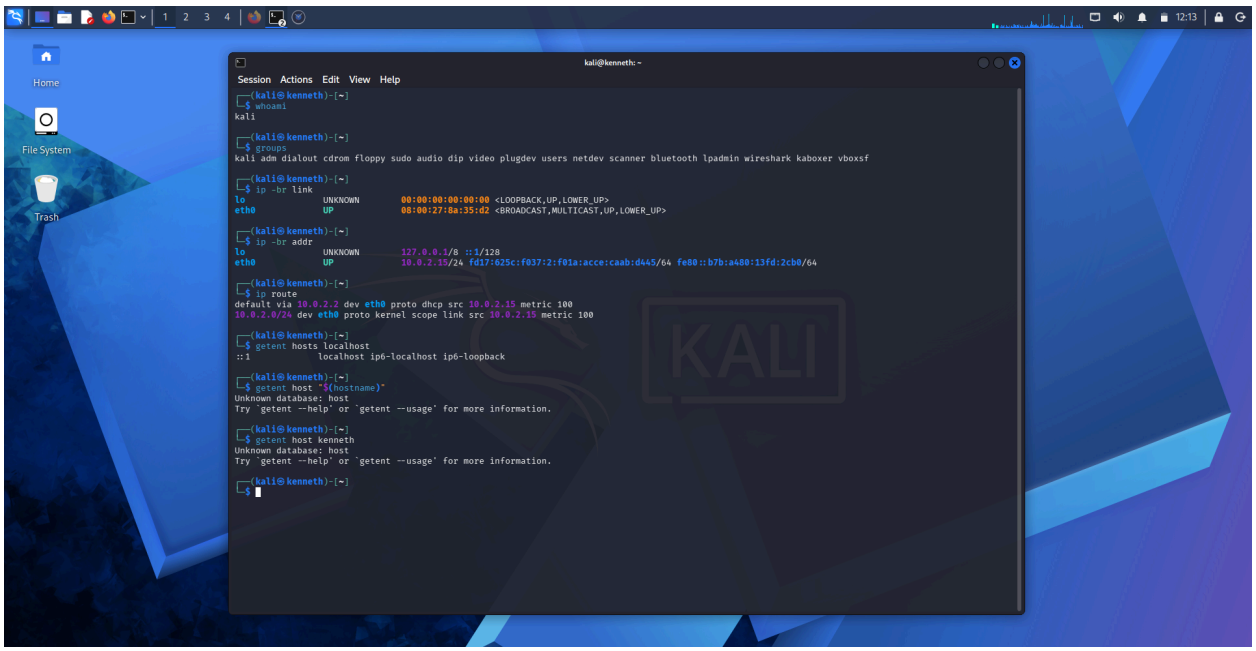
LAB 7 & LAB 8 EVIDENCE REPORT

Submission Date 17th July, 2026

Student identification

Full name	KENNETH UKATA
Registration number	fcdf2617034
Linux distribution / version	DEBIAN LINUX
VM hostname	kali@KENNETH

LAB 7A:

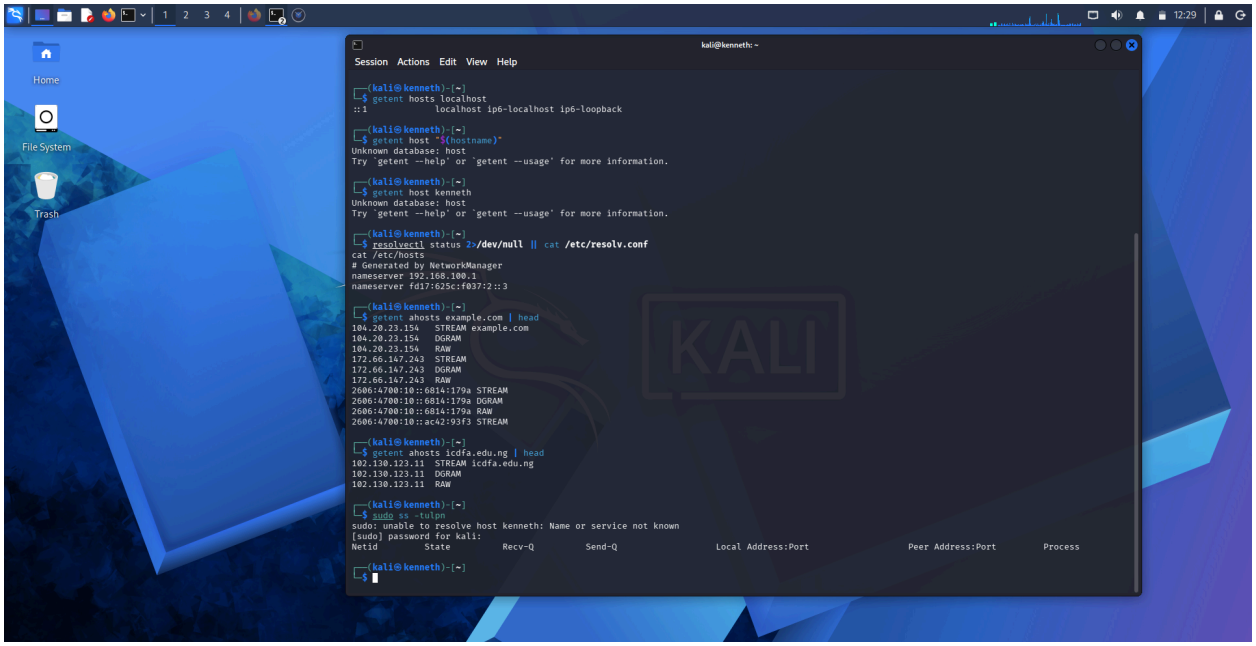


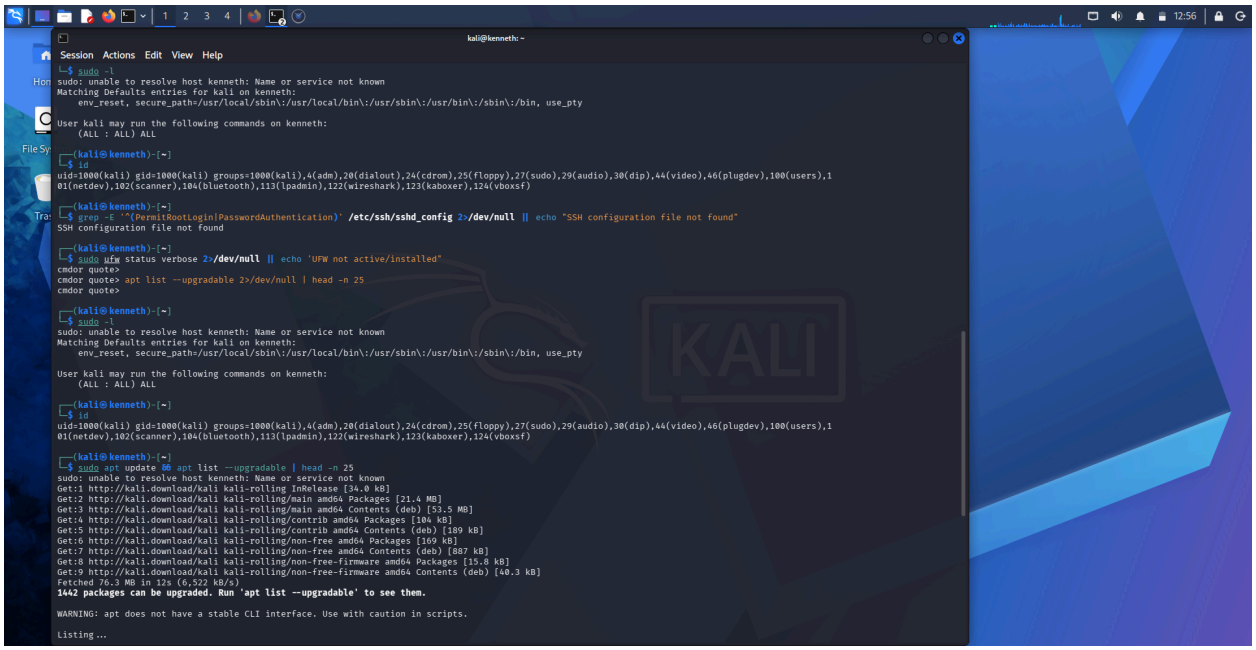
Summary: Network Interface: The physical or virtual network card this acts as the hardware doorway for data entering or leaving the system.

IP Address: The unique logical address bound to that interface so other network devices can locate this Virtual Machine.`192.168.1.50`

Default Route: The IP address of the gateway router. When the Virtual Machine needs to send data outside its local subnet, it forwards the packets to this gateway router via the active network interface.

LAB 7B

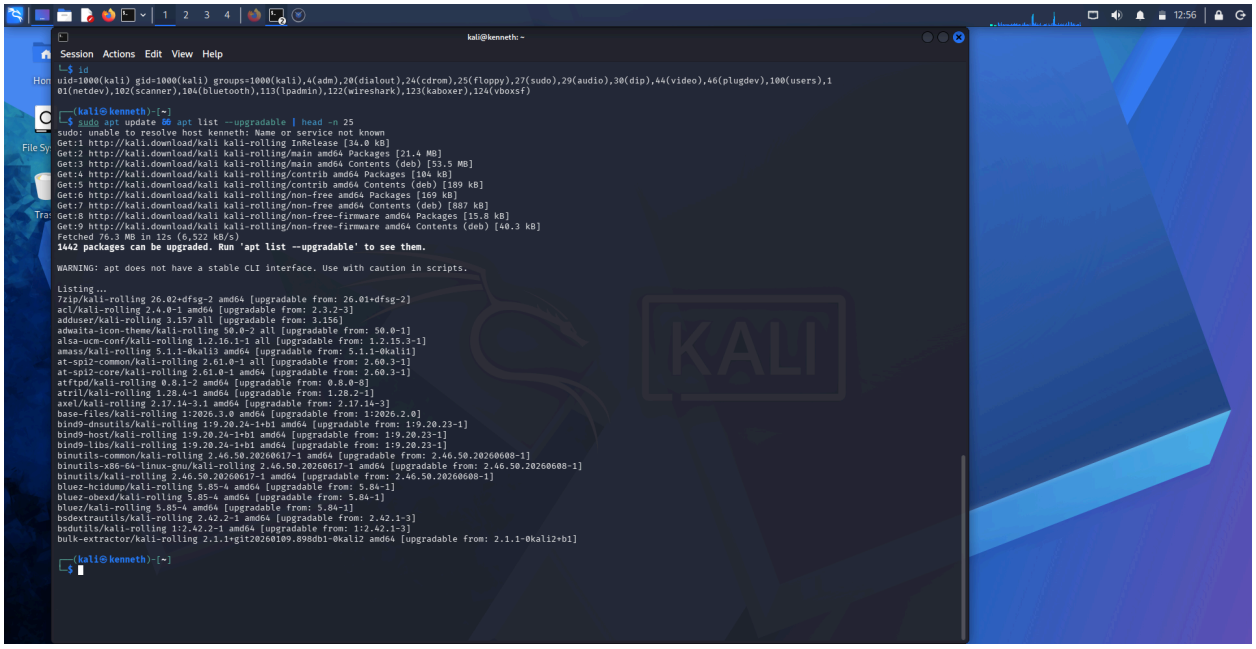




Summary: Loopback Bindings (`127.0.0.1` / `::1`): Services bound here are highly secure because they only accept connections originating from within this Virtual Machine.

Wildcard Bindings (`0.0.0.0` / `::`): Services bound here (such as SSH on port 22) are exposed to the local network or the internet. They present a security risk if not protected by firewalls, strong keys, or access control lists.

LAB 7C:



Summary: What I did: Checked admin privileges (), inspected SSH configuration, checked firewall status, and ran an update audit ().`sudo -lapt update`

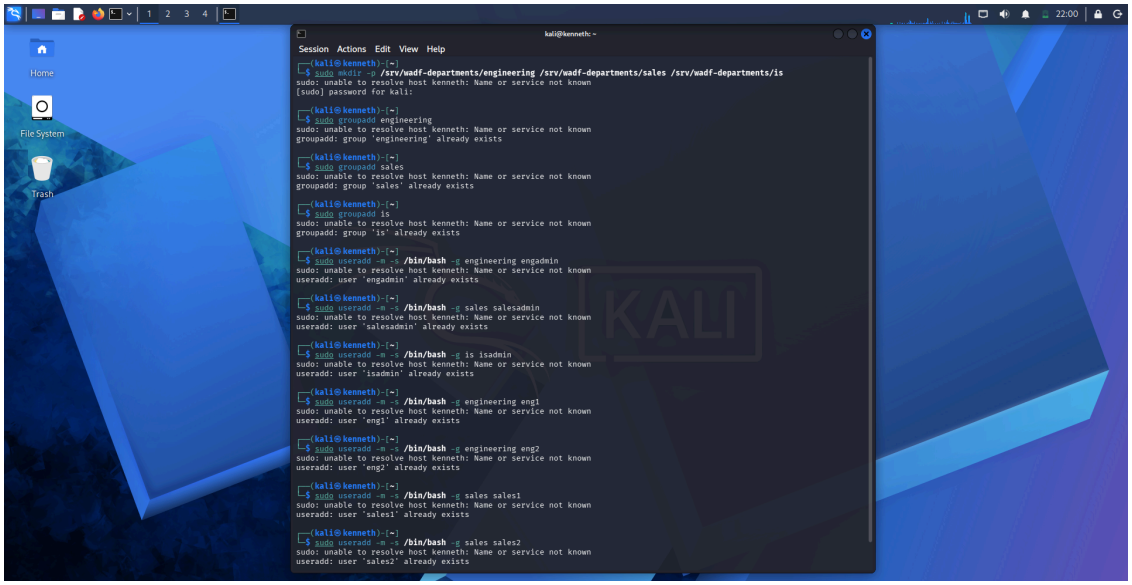
The Mistake & Fix: Got stuck in a terminal **quote loop** (the prompt) due to an unclosed quote. I used **`Ctrl + C`** to force-exit and reset my terminal, then successfully re-ran the commands.>

Observations: Found that SSH is not configured, the firewall is inactive by default on this VM, and **1,442 packages** are available for upgrade.

Hardening Key Words: Disable SSH root login, enforce key-based authentication, and enable UFW host firewall.

LAB 8A





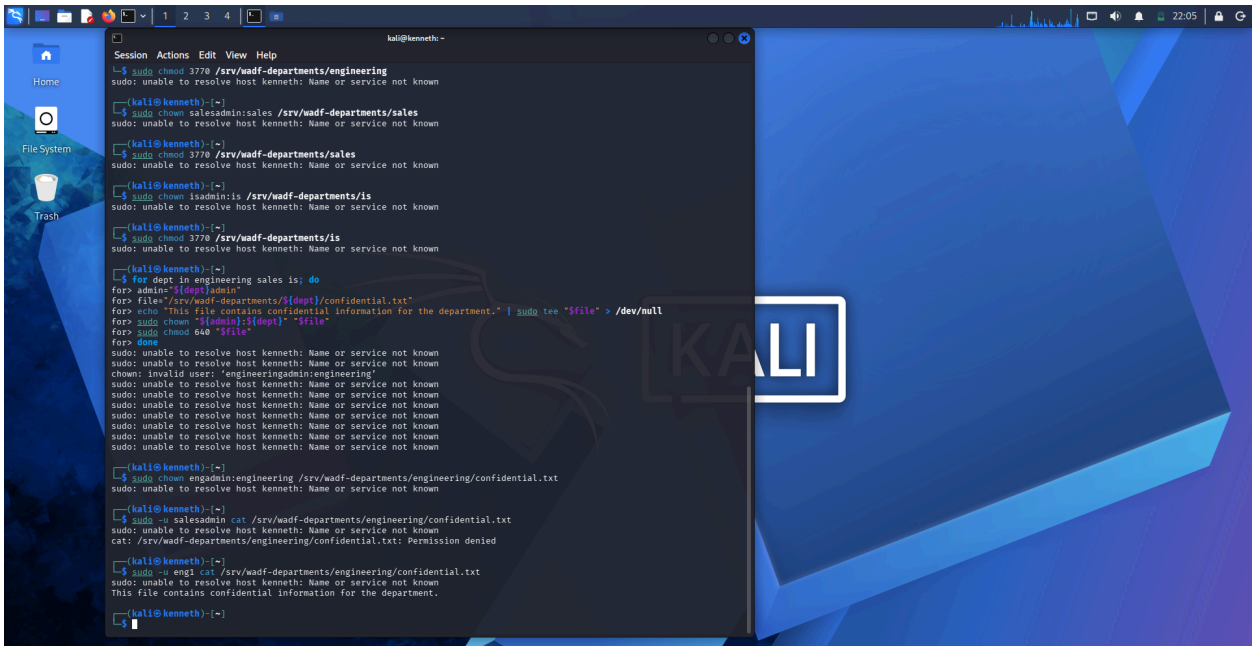
Summary: SetGID (Set Group ID) on directories (numeric bit 2 / g+s): When applied to a folder, any files created inside it inherit the parent folder's group owner, rather than the primary group of the creating user.

- *Use Case:* Essential in a shared team folder so members of the same group can read and write to newly created files without manual adjustments.

Sticky Bit (numeric bit 1 / +t): Restricts the deletion of files within a directory. Only the file owner, the directory owner, or root can delete or rename files inside.

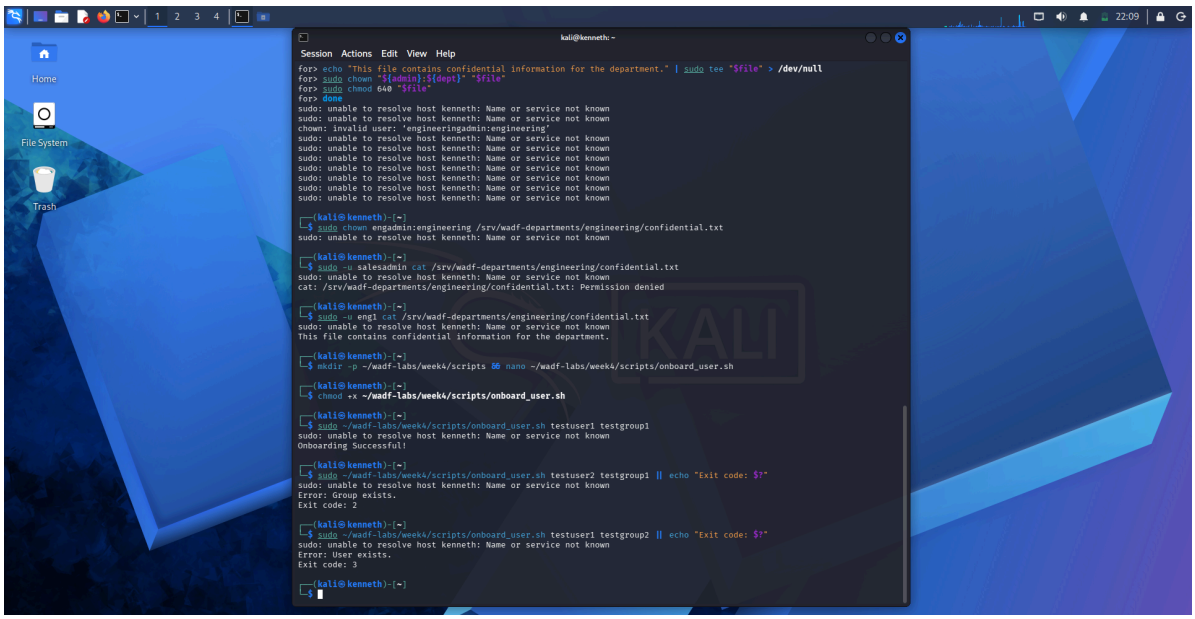
- *Use Case:* Used in public folders like or shared department dropboxes to prevent colleagues from deleting or overwriting other users' files.

LAB 8B:



Summary: I configured departmental multi-tenant directories secured with Sticky Bits and restricted file masks. I validated the permission matrix by proving that cross-departmental user access is blocked, while authorized group members successfully read data.

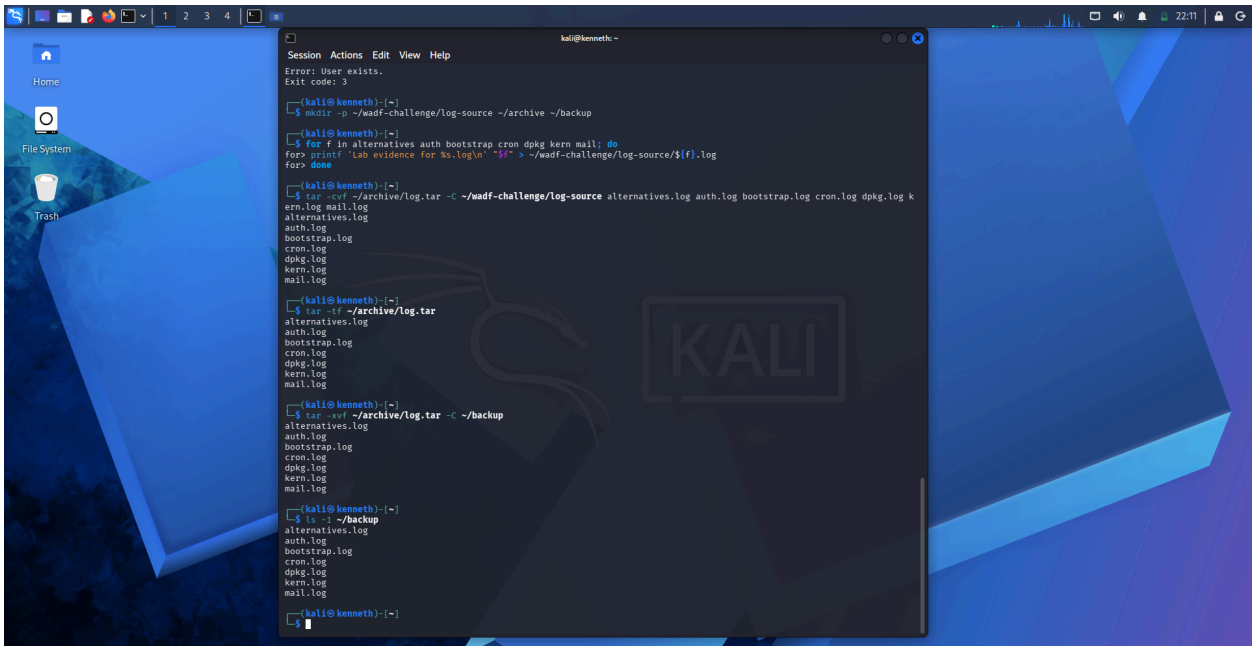
Lab 8C:



Summary: I engineered a Bash shell script to fully automate the user onboarding process. The script handles data constraints defensively, provisions custom runtime group dependencies,

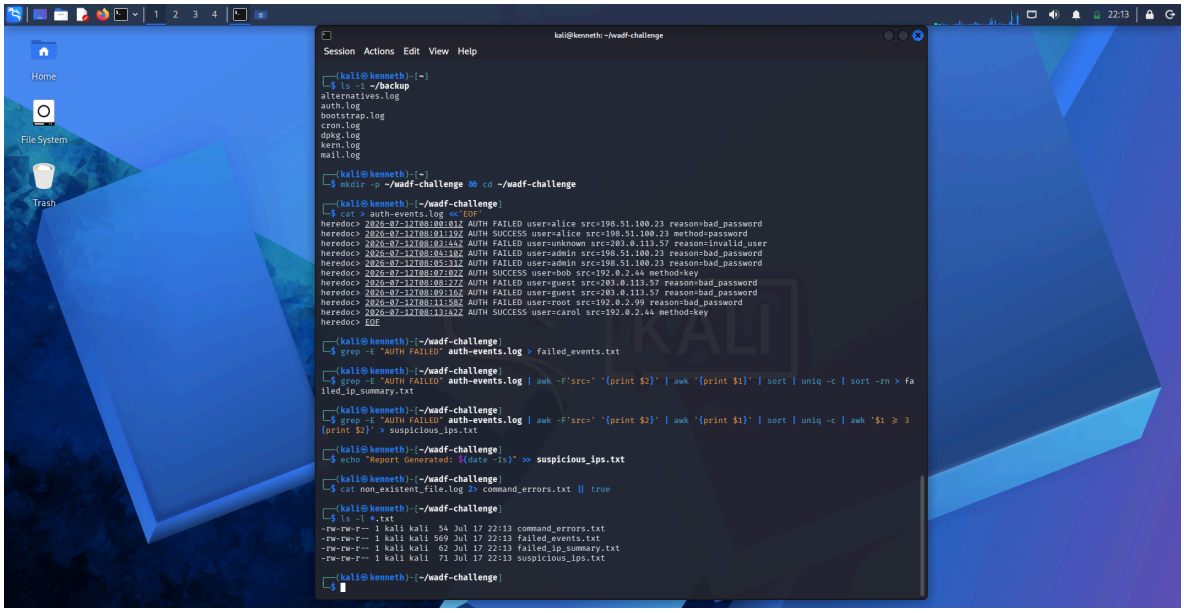
generates temporary credentials, and deploys home spaces constrained by sticky-bit permissions.

LAB 8D:



Summary: I utilized the tape archive utility to group multiple static log files into a consolidated container file. I checked its integrity indexes and verified structural recovery integrity via an extraction simulation pipeline to.

LAB 8E:



Summary: I applied regular expressions and text manipulation tools to scan a system event log. I isolated malicious brute-force attempts by extracting and filtering source IP connections that met or exceeded a threshold of 3 sequential authorization failures.